



PERSONAL INFO

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GITHUB
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LOCATION
Wrocław, Poland

SKILLS

ML & Computer Vision: PyTorch, TensorFlow/Keras, YOLOv26 (Ultralytics), OpenCV, scikit-learn, statsmodels · object & anomaly detection, time-series forecasting (CNN, LSTM/GRU)

Programming & Data: Python, SQL, NumPy, Git/GitHub, Docker, CI/CD, Flask, REST, SQLite, n8n

Real-Time Systems: multithreaded inference pipelines, GPU/CUDA, real-time video streaming, edge deployment

Industrial AI / OT: OPC UA, MQTT, Modbus, PLC (TIA Portal/CODESYS), HMI/SCADA, industrial sensors/IoT, Node-Red

Other: Driving License (Cat. B), SEP Qualification (up to 1 kV)

LANGUAGES

Polish	Native
English	B2
German	A2

COURSES & CERTIFICATION

Machine Learning Course

Python in Industrial Automation

Programming Siemens S7-1200 / S7-1500 PLCs using Structured Control Language (SCL)

Grzegorz Czyrniański

Machine Learning & Computer Vision Engineer

Application Developer at JUMO with an M.Sc. in IT Automation Systems and hands-on experience executing industrial projects end-to-end — from PLC programming and system design to commissioning. I bridge traditional Operational Technology (PLC, industrial networks, HMI/SCADA) with modern data-driven methods, and I am now focused on moving into machine learning and computer vision. My goal is to design and deploy ML/CV solutions — particularly object detection, anomaly detection and predictive maintenance — for complex Industry 4.0 challenges.

PROJECTS

Real-Time Object Detection & Sorting System · Computer Vision + PLC

- Built an end-to-end real-time CV pipeline (Python, YOLOv8/Ultralytics, PyTorch + CUDA, OpenCV) detecting and classifying products from a live camera at **30 FPS** on GPU.
- Built it with a modular, multithreaded architecture so real-time inference and the live web dashboard run in parallel without blocking each other.
- Integrated vision with industrial control over Modbus TCP to a JUMO variTRON 300 PLC (Structured Text), driving a Factory I/O digital-twin line that sorts detected objects in real time.
- Delivered a Flask dashboard (live MJPEG + real-time stats) with SQLite event logging, packaged with Docker and covered by unit tests.

M.Sc. Thesis — Temperature Forecasting System · Classical & Deep Learning

- Benchmarked classical models (ARIMA/SARIMA) against neural networks (LSTM, GRU, CNN and hybrids) for 1-hour and 24-hour-ahead time-series forecasting in TensorFlow (GPU).
- Owned the full ML workflow — preprocessing, model selection and evaluation (MAE / RMSE / R²) on a real 2023–2024 test set.
- Deployed the best model as a real-time predictive app integrated with a PT1000 sensor and PLC over OPC UA, with SQLite archiving (Industry 4.0).

Multi-Agent AI Automation System · LLM + Multimodal

- Built an autonomous multi-agent workflow in n8n using LLMs (Google Gemini, Perplexity) with multimodal pipelines for image/audio analysis and automated web research.
- Integrated Telegram and Google Workspace APIs to ship a working product for user interaction and automated task scheduling.

EXPERIENCE

Application Developer

JUMO, Wrocław

2025-08 – present

- Design and commission end-to-end automation and data-acquisition solutions with international engineering teams (incl. the Austrian branch).
- Program and optimize PLC controllers (Siemens TIA Portal, JUMO variTRON / CODESYS, Structured Text) for reliable control of complex industrial processes.
- Build industrial data pipelines and cross-platform communication (OPC UA, MQTT, Modbus, EtherNet/IP) feeding real-time monitoring and IoT-ready infrastructure.
- Design HMI/SCADA visualization connectivity for remote monitoring, alarming and data archiving.

Inside Sales

JUMO, Wrocław

2024-06 – 2025-08

- Technical selection of process-automation and sensing components; interface between customers, production and subcontractors.

Internship – Electrical Installation & Measurement

Electro, Olawa

2023 – 2024-05

- Electrical installation and interpretation of technical diagrams and documentation.

EDUCATION

M.Sc., IT Automation Systems

Wrocław University of Science and Technology

Specialization: Intelligent Industry Systems 4.0

2024-03 – 2025-07

B.Eng., Industrial Automation

Wrocław University of Science and Technology

Specialization: Automation of Machines, Vehicles and Devices

2021-10 – 2024-02

